Chapter 5: Joint Distributions of Random Variables

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CSAI2121 Probability and Statistics
Macao Polytechnic University

Learning Objectives

By the end of this lecture, you should be able to:

- Define and interpret the **joint PMF** and **joint PDF** of two random variables
- Derive **marginal** distributions from a joint distribution
- Compute **conditional** distributions from joint and marginal distributions
- Evaluate **iterated integrals** to compute probabilities from a continuous joint PDF
- Determine whether two random variables are **independent**
- Recognize joint distributions in AI/ML contexts: confusion matrices, sensor fusion, Naive Bayes

Discrete Joint Distributions

Motivation: Why Joint Distributions?

- So far: one random variable at a time (Chapters 3–4).
- Many real questions involve **two or more** random variables together.
- Examples: a model's *predicted* label vs. its *true* label; temperature vs. humidity from two sensors; two features in a dataset (X_1, X_2).
- We need tools to describe how two random variables behave **jointly** — not just separately.

Recall: A Single Random Variable

- Discrete: PMF $p_X(x) = P(X = x)$, with $\sum_x p_X(x) = 1$.
- Continuous: PDF $f_X(x)$, with $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$.

Today's Extension

We now consider **two** random variables X and Y *simultaneously*, and ask about $P(X = x, Y = y)$.

Two Random Variables: Basic Setup

- Let X and Y be two random variables defined on the same experiment.
- Their **joint behavior** is described by the probability of both events occurring together:

$$P(X = x,\ Y = y) \quad \text{(discrete)},$$

$$P(a \le X \le b,\ c \le Y \le d) \quad \text{(continuous)}.$$

- Today we build the discrete case first, then extend to the continuous case.

Joint PMF (Discrete Case): Definition

Definition

The **joint probability mass function** of discrete random variables X and Y is

$$p_{X,Y}(x, y) = P(X = x, \ Y = y)$$

- Property 1: $p_{X,Y}(x, y) \geq 0$ for all (x, y)
- Property 2: $\sum_x \sum_y p_{X,Y}(x, y) = 1$

Joint PMF Example: Table Representation

Let $X =$ device type ($0 =$ Mobile, $1 =$ Desktop), $Y =$ subscription plan ($0 =$ Free, $1 =$ Premium).

	$Y = 0$ (Free)	$Y = 1$ (Premium)
$X = 0$ (Mobile)	0.30	0.20
$X = 1$ (Desktop)	0.15	0.35

Joint PMF Example: Table Representation

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	$Y = 0$ (Free)	$Y = 1$ (Premium)
$X = 0$ (Mobile)	0.30	0.20
$X = 1$ (Desktop)	0.15	0.35

Check: $0.30 + 0.20 + 0.15 + 0.35 = 1.00$ ✓

From Joint PMF to Marginal PMF

Definition

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

- "Marginal" = sum *across the other variable* — literally the row/column totals in a table.
- This recovers the single-variable PMF of X (or Y) alone, ignoring the other.

Marginal PMF Example (Continuing Slide 7)

	$Y=0$	$Y=1$	$p_X(x)$
$X=0$	0.30	0.20	**0.50**
$X=1$	0.15	0.35	**0.50**
$p_Y(y)$	**0.45**	**0.55**	1.00

Marginal PMF Example (Continuing Slide 7)

	$Y=0$	$Y=1$	$p_X(x)$
$X=0$	0.30	0.20	**0.50**
$X=1$	0.15	0.35	**0.50**
$p_Y(y)$	**0.45**	**0.55**	1.00

- $p_X(0) = 0.50$: half of users are on mobile.
- $p_Y(1) = 0.55$: 55% of users are on the premium plan.

AI Application: A Confusion Matrix *Is* a Joint PMF

- A classifier's **confusion matrix** counts pairs (predicted label $\hat{Y}$, true label Y).
- Dividing by the total sample size n turns counts into an empirical **joint PMF**:

$$p_{\hat{Y},Y}(\hat{y}, y) = \frac{\text{count}(\hat{Y} = \hat{y},\ Y = y)}{n}$$

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Example ($n = 100$, binary classifier):

	$Y = 0$	$Y = 1$	Row sum
$\hat{Y} = 0$	45	5	50
$\hat{Y} = 1$	10	40	50
Col sum	55	45	100

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Example ($n = 100$, binary classifier):

	$Y = 0$	$Y = 1$	Row sum
$\hat{Y} = 0$	45	5	50
$\hat{Y} = 1$	10	40	50
Col sum	55	45	100

$p_{\hat{Y},Y}(1,1) = 0.40, p_Y(1) = 0.45, p_{\hat{Y}|Y}(1 \mid 1) = \dfrac{0.40}{0.45} \approx 0.889 =$ Recall

Conditional PMF (Discrete Case)

Definition

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \qquad \text{provided } p_Y(y) > 0$$

- Answers: "Given that $Y = y$, what is the distribution of X?"
- Same idea as conditional probability $P(A \mid B) = P(A \cap B)/P(B)$ from earlier chapters.

Conditional PMF Example (Continuing Slide 7)

Given a user is on the **Premium** plan ($Y = 1$), what is the distribution of device type?

Conditional PMF Example (Continuing Slide 7)

Given a user is on the **Premium** plan ($Y = 1$), what is the distribution of device type?

$$p_{X|Y}(0 \mid 1) = \frac{p_{X,Y}(0,1)}{p_Y(1)} = \frac{0.20}{0.55} \approx 0.364$$

$$p_{X|Y}(1 \mid 1) = \frac{p_{X,Y}(1,1)}{p_Y(1)} = \frac{0.35}{0.55} \approx 0.636$$

Conditional PMF Example (Continuing Slide 7)

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$$p_{X|Y}(1 \mid 1) = \frac{p_{X,Y}(1,1)}{p_Y(1)} = \frac{0.35}{0.55} \approx 0.636$$

Check: $0.364 + 0.636 = 1.00$ ✓ — Premium users are more likely to be on desktop.

Relationship: Joint, Marginal, Conditional

Key Identity

$$p_{X,Y}(x, y) = p_{X|Y}(x \mid y) \cdot p_Y(y) = p_{Y|X}(y \mid x) \cdot p_X(x)$$

- This is simply the multiplication rule from Chapter 2, restated for random variables.
- **Joint** describes both variables together; **marginal** describes one alone; **conditional** describes one *given* the other.

Segment Summary: Discrete Joint Distributions

- Joint PMF: $p_{X,Y}(x, y) = P(X = x, Y = y)$
- Marginal PMF: sum the joint PMF over the *other* variable
- Conditional PMF: divide joint by the relevant marginal
- Next: we extend all three ideas to **continuous** random variables, where sums become integrals.

Continuous Joint Distributions and Iterated Integrals

From Discrete to Continuous: Motivation

- Just as in Chapter 4, many pairs of variables are continuous: two sensor readings, two continuous model outputs, two physical measurements.
- As before: $P(X = x, Y = y) = 0$ for any exact pair — we need a **density**.
- We now need a joint **PDF**, and a way to integrate over *two* variables.

Joint PDF: Definition and Properties

Definition

The **joint probability density function** $f_{X,Y}(x, y)$ satisfies

$$P\big((X, Y) \in A\big) = \iint_A f_{X,Y}(x, y)\, dx\, dy$$

- Property 1: $f_{X,Y}(x, y) \geq 0$
- Property 2: $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$
- As in Chapter 4, $f_{X,Y}$ itself is not a probability — only *volumes* under it are.

Mathematical Tool: The Iterated Integral

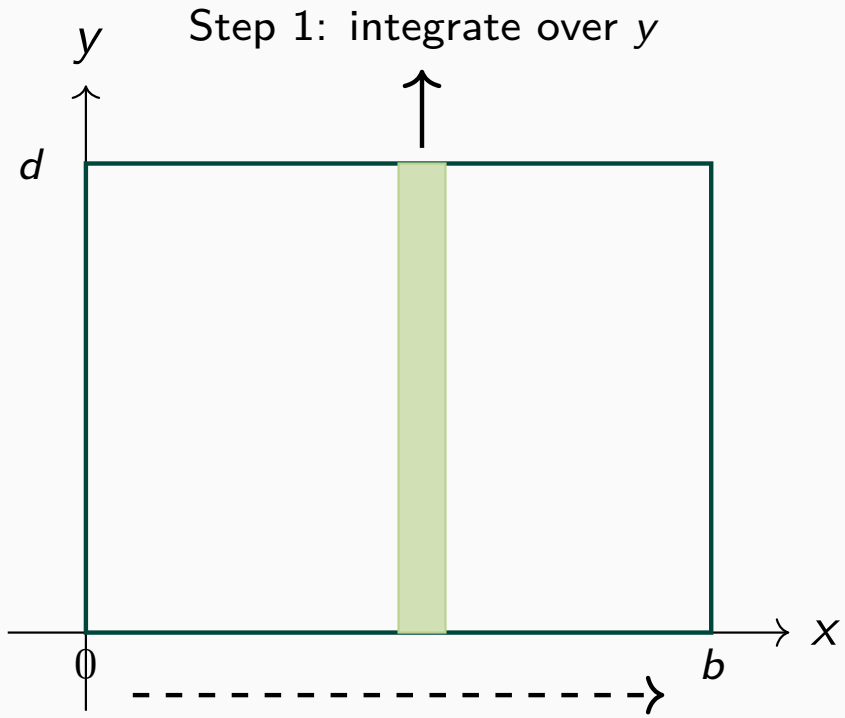
Key Idea

A double integral over a rectangular region is computed as **two single-variable integrals**, one after another:

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \int_a^b \left[\int_c^d f(x, y)\, dy \right] dx$$

- Step 1: integrate over y, treating x as a **constant** — this gives a function of x alone.
- Step 2: integrate that resulting function over x.
- This is *not* new mathematics — it is ordinary single-variable integration, applied twice.

Visualizing the Iterated Integral



- The shaded strip at a fixed x is integrated over y first.
- Repeating this for every x and accumulating gives the double integral.

Worked Example 1: Product-Form Joint PDF

Let $f_{X,Y}(x, y) = c\,xy$ for $0 \le x \le 2,\ 0 \le y \le 1$.

Worked Example 1: Product-Form Joint PDF

Let $f_{X,Y}(x,y) = c\,xy$ for $0 \le x \le 2,\ 0 \le y \le 1$. **Step 1: find c.**

$$\int_0^2 \int_0^1 c\,xy\,dy\,dx = c\int_0^2 x\left[\frac{y^2}{2}\right]_0^1 dx = c\int_0^2 \frac{x}{2}\,dx = c(1) = 1 \ \Rightarrow\ c = 1$$

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Step 2: find $P(1 \le X \le 2,\ 0 \le Y \le 0.5)$.

$$\int_1^2 \int_0^{0.5} xy\,dy\,dx = \int_1^2 x(0.125)\,dx = 0.125\left[\frac{x^2}{2}\right]_1^2 = 0.125(1.5) = 0.1875$$

Worked Example 2: Sum-Form Joint PDF (Harder)

Let $f_{X,Y}(x, y) = c(x + y)$ for $0 \le x \le 1,\ 0 \le y \le 2$.

Worked Example 2: Sum-Form Joint PDF (Harder)

Let $f_{X,Y}(x,y) = c(x+y)$ for $0 \le x \le 1,\ 0 \le y \le 2$. **Step 1: find c.**

$$\int_0^1 \int_0^2 c(x+y)\,dy\,dx = c\int_0^1 (2x+2)\,dx = c(1+2) = 3c = 1 \ \Rightarrow\ c = \tfrac{1}{3}$$

Worked Example 2: Sum-Form Joint PDF (Harder)

Let $f_{X,Y}(x,y) = c(x+y)$ for $0 \le x \le 1,\ 0 \le y \le 2$. **Step 1: find c.**

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Step 2: find $P(0 \le X \le 0.5,\ 1 \le Y \le 2)$.

$$\int_0^{0.5} \int_1^2 \tfrac{1}{3}(x+y)\,dy\,dx = \tfrac{1}{3}\int_0^{0.5} (x+1.5)\,dx = \tfrac{1}{3}(0.125+0.75) = \tfrac{7}{24} \approx 0.29$$

Worked Example 2: Sum-Form Joint PDF (Harder)

Let $f_{X,Y}(x,y) = c(x+y)$ for $0 \le x \le 1,\ 0 \le y \le 2$. **Step 1: find c.**

$$\int_0^1 \int_0^2 c(x+y)\,dy\,dx = c\int_0^1 (2x+2)\,dx = c(1+2) = 3c = 1 \ \Rightarrow\ c = \tfrac{1}{3}$$

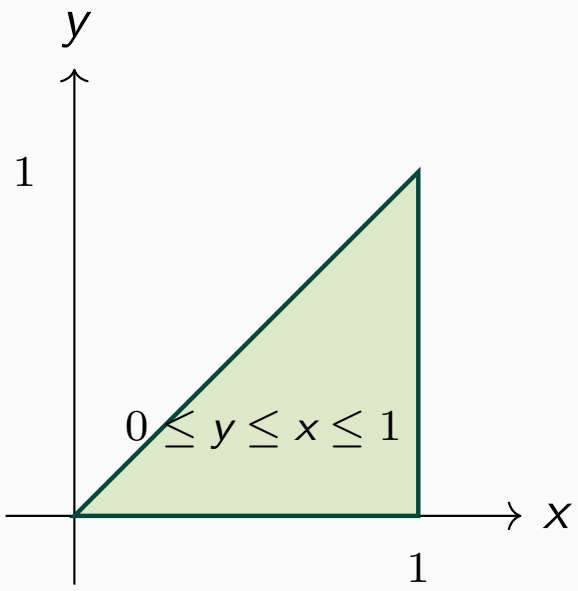
Step 2: find $P(0 \le X \le 0.5,\ 1 \le Y \le 2)$.

$$\int_0^{0.5} \int_1^2 \tfrac{1}{3}(x+y)\,dy\,dx = \tfrac{1}{3}\int_0^{0.5} (x+1.5)\,dx = \tfrac{1}{3}(0.125+0.75) = \tfrac{7}{24} \approx 0.29$$

Note

Unlike Example 1, $x+y$ cannot be split into "a function of x" times "a function of y". Keep this in mind for later.

Extension: Non-Rectangular Support Regions



- If the support is triangular (e.g., $0 \le y \le x \le 1$), the integration **limits** depend on the other variable:

$$\int_0^1 \left[\int_0^x f(x, y)\, dy \right] dx$$

- **Not required** for this course's assessments — shown here for completeness.

Marginal PDF (Continuous Case)

Definition

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx$$

- Same idea as the discrete case: integrate out the variable you don't care about.

Marginal PDF Example (Continuing Worked Example 1)

Recall $f_{X,Y}(x, y) = xy$ on $0 \leq x \leq 2,\ 0 \leq y \leq 1$.

Marginal PDF Example (Continuing Worked Example 1)

Recall $f_{X,Y}(x, y) = xy$ on $0 \leq x \leq 2,\ 0 \leq y \leq 1$.

$$f_X(x) = \int_0^1 xy\, dy = \frac{x}{2}, \qquad 0 \leq x \leq 2$$

$$f_Y(y) = \int_0^2 xy\, dx = 2y, \qquad 0 \leq y \leq 1$$

Marginal PDF Example (Continuing Worked Example 1)

Recall $f_{X,Y}(x,y) = xy$ on $0 \le x \le 2,\ 0 \le y \le 1$.

$$f_X(x) = \int_0^1 xy\,dy = \frac{x}{2}, \qquad 0 \le x \le 2$$

$$f_Y(y) = \int_0^2 xy\,dx = 2y, \qquad 0 \le y \le 1$$

Check: $\int_0^2 \frac{x}{2}dx = 1, \quad \int_0^1 2y\,dy = 1$ ✓

Marginal PDF Example (Continuing Worked Example 1)

Recall $f_{X,Y}(x,y) = xy$ on $0 \le x \le 2,\ 0 \le y \le 1$.

$$f_X(x) = \int_0^1 xy\,dy = \frac{x}{2}, \qquad 0 \le x \le 2$$

$$f_Y(y) = \int_0^2 xy\,dx = 2y, \qquad 0 \le y \le 1$$

Check: $\int_0^2 \frac{x}{2}dx = 1, \quad \int_0^1 2y\,dy = 1$ ✓

Observation (keep this in mind)

$$f_X(x) \cdot f_Y(y) = \frac{x}{2} \cdot 2y = xy = f_{X,Y}(x,y)$$

AI Application: Sensor Fusion and Anomaly Detection

- IoT and monitoring systems often track two continuous signals jointly, e.g., temperature X and humidity Y.
- Given a joint PDF $f_{X,Y}(x, y)$ learned from normal operation, define an anomaly as:

$$\text{Anomaly} \iff f_{X,Y}(x, y) < \tau \quad (\tau = \text{threshold})$$

- Even if X alone looks normal and Y alone looks normal, the **combination** (x, y) can still be anomalous.

Key Point

This is exactly why marginal distributions alone are insufficient — the *joint* density captures relationships that marginals cannot.

Instructor Demo 1: A Joint PDF as a Surface

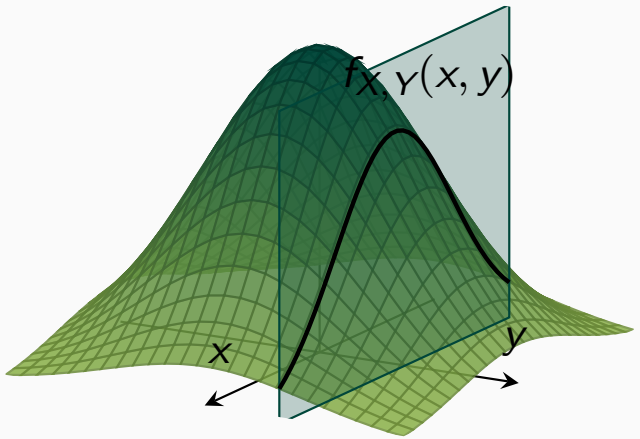
- A joint PDF $f_{X,Y}(x, y)$ can be visualized as a **surface** over the (x, y)-plane; probability corresponds to the **volume** under the surface above a region.
- Contour lines (constant-height slices) show where the density is high or low — similar to a topographic map.
- Compare with Chapter 4: there, probability was an *area*; now it is a *volume*.

Conditional PDF (Continuous Case)

Definition

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \qquad \text{provided } f_Y(y) > 0$$

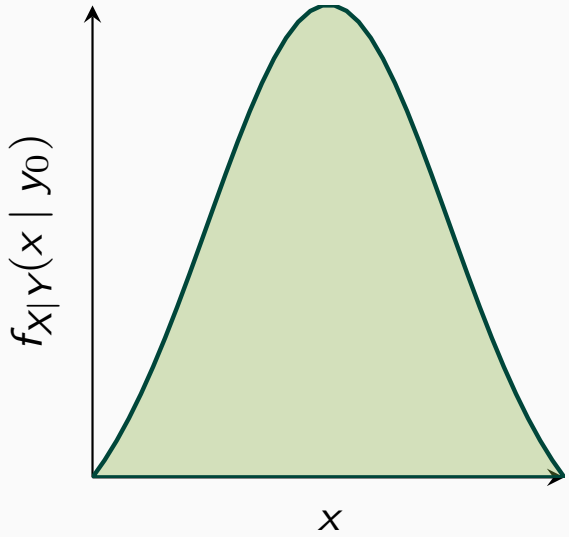
- Same structure as the discrete case, with sums replaced by integrals.



slice at $y = y_0$

$\Rightarrow$

normalize by $f_Y(y_0)$



Conditional PDF Example (Continuing Worked Example 1)

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{xy}{2y} = \frac{x}{2}, \qquad 0 \le x \le 2$$

Conditional PDF Example (Continuing Worked Example 1)

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{xy}{2y} = \frac{x}{2}, \qquad 0 \le x \le 2$$

Key Observation

The result does **not depend on** y: $f_{X|Y}(x \mid y) = f_X(x)$ for every y. This is an early sign of **independence** — formalized in the next segment.

Segment Summary: Continuous Joint Distributions

- Joint PDF: probability = volume under $f_{X,Y}(x, y)$ over a region
- Iterated integral: two ordinary single-variable integrals, done in sequence
- Marginal PDF: integrate out the other variable
- Conditional PDF: divide joint by the relevant marginal
- Next: when does $f_{X|Y}(x \mid y) = f_X(x)$ hold for *all* y? $\rightarrow$ Independence

Independence of Random Variables

Independence of Random Variables: Definition

Definition

X and Y are **independent** if, for *all* x, y:

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y) \quad \text{(discrete)}$$

$$f_{X,Y}(x, y) = f_X(x)\, f_Y(y) \quad \text{(continuous)}$$

- Equivalently: $p_{X|Y}(x \mid y) = p_X(x)$ (or the continuous analogue) for all y.
- Intuition: knowing Y gives **no information** about X.

Independence Check: Continuous Case — Two Contrasting Examples

Example 1 ($f = xy$, from Slide 19/23):

$$f_X(x) f_Y(y) = \frac{x}{2} \cdot 2y = xy = f_{X,Y}(x, y) \;\Rightarrow\; \textbf{Independent}$$

Independence Check: Continuous Case — Two Contrasting Examples

Example 1 ($f = xy$, from Slide 19/23):

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Example 2 ($f = \frac{1}{3}(x+y)$, from Slide 20): marginals are

$$f_X(x) = \frac{2x+2}{3}, \quad f_Y(y) = \frac{0.5+y}{3}$$

At $x = 0, y = 0$:

$$f_X(0)f_Y(0) = \frac{2}{3} \cdot \frac{0.5}{3} = \frac{1}{9} \approx 0.111 \neq 0 = f_{X,Y}(0,0) \;\Rightarrow\; \textbf{Not Independent}$$

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Teaching Point

One counterexample is enough to disprove independence; proving independence requires the equality to hold *everywhere*.

Independence Check: Discrete Case 1 (AI Context, Not Independent)

X = model architecture (0 = CNN, 1 = Transformer), Y = achieves high accuracy (0 = No, 1 = Yes).

	$Y=0$	$Y=1$	Row sum
$X=0$ (CNN)	0.25	0.35	0.60
$X=1$ (Transformer)	0.15	0.25	0.40
Col sum	0.40	0.60	1.00

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$$p_X(1)\,p_Y(1) = 0.40 \times 0.60 = 0.24 \;\neq\; 0.25 = p_{X,Y}(1,1)$$

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$$p_X(1)\,p_Y(1) = 0.40 \times 0.60 = 0.24 \;\neq\; 0.25 = p_{X,Y}(1,1)$$

Not independent — architecture is associated with achieving high accuracy. Note how close 0.24 and 0.25 are: independence must be checked *exactly*, not approximately.

Independence Check: Discrete Case 2 (Independent, by Construction)

$X =$ server region (0 = US, 1 = EU), $Y =$ request outcome (0 = Fail, 1 = Success), with $p_X(0) = 0.6, p_X(1) = 0.4$ and $p_Y(0) = 0.3, p_Y(1) = 0.7$, constructed to be independent:

	$Y = 0$	$Y = 1$	Row sum
$X = 0$ (US)	0.18	0.42	0.60
$X = 1$ (EU)	0.12	0.28	0.40
Col sum	0.30	0.70	1.00

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$p_X(1)\, p_Y(1) = 0.4 \times 0.7 = 0.28 = p_{X,Y}(1,1)$ ✓ (holds for every cell)

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$p_X(1)\,p_Y(1) = 0.4 \times 0.7 = 0.28 = p_{X,Y}(1,1)$ ✓ (holds for every cell)

Independent — server region does not affect the success rate of requests, a desirable property in system design.

Instructor Demo 2: From Independence Toward Correlation

- Independence is a strict **yes/no** property: either the equality holds for all (x, y), or it does not.
- But in Slide 30, the CNN/Transformer example was "nearly" independent (0.24 vs. 0.25) — how strong is the association?
- We currently have no way to measure the **strength** or **direction** of a relationship — only whether one exists.
- This motivates **covariance** and **correlation**, coming next week.

Common Misconceptions about Independence

- Checking **one** pair (x, y) is enough to *disprove* independence, but **all** pairs must be checked to *prove* it.
- $p_{X,Y}(x, y) = 0$ does **not** automatically mean "not independent" — compare it against $p_X(x)p_Y(y)$.
- Independence is the **same definition** for discrete and continuous cases — only sums vs. integrals differ.
- (Preview) Zero correlation does **not** imply independence — more on this next week.

Summary Table: Discrete vs. Continuous

	Discrete	Continuous
Joint	$p_{X,Y}(x,y) = P(X=x, Y=y)$	$f_{X,Y}(x,y),\ P(A) = \iint_A f_{X,Y}\,dx\,dy$
Marginal	$p_X(x) = \sum_y p_{X,Y}(x,y)$	$f_X(x) = \int f_{X,Y}(x,y)\,dy$
Conditional	$p_{X\|Y}(x\|y) = \dfrac{p_{X,Y}(x,y)}{p_Y(y)}$	$f_{X\|Y}(x\|y) = \dfrac{f_{X,Y}(x,y)}{f_Y(y)}$
Independence	$p_{X,Y} = p_X\,p_Y$ (all x,y)	$f_{X,Y} = f_X\,f_Y$ (all x,y)

AI Application: The Naive Bayes Independence Assumption

Naive Bayes assumes features are **conditionally independent** given the class:

$$p(x_1, \ldots, x_d \mid y) = \prod_{i=1}^{d} p(x_i \mid y)$$

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Example (spam classification): given $Y = \text{spam}$, assume "contains 'free'" (X_1) and "contains 'winner'" (X_2) are independent given spam:

$$p(X_1 = 1, X_2 = 1 \mid \text{spam}) = p(X_1 = 1 \mid \text{spam}) \cdot p(X_2 = 1 \mid \text{spam}) = 0.7 \times 0$$

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Caveat

This is an *assumption*, not always true in real data — it drastically simplifies computation, hence "naive".

AI Application: Recommendation Systems and Looking Ahead

- Let $X =$ a user's rating of Product A, $Y =$ rating of Product B (e.g., a phone and a matching case).
- If the products are complementary, ratings are likely **not independent** — liking A tends to go with liking B.
- Right now we can only say **yes/no** to independence via $f_{X,Y} = f_X f_Y$.
- **Next week's question:** if not independent, how *strong* and in what *direction* is the relationship?
- Answer: **Covariance** and the **Correlation Coefficient**.

Class Exercise

Problem

1. Let $f_{X,Y}(x, y) = k x^2 y$ for $0 \leq x \leq 1,\ 0 \leq y \leq 2$. Find k and compute $P(X < 0.5,\ Y < 1)$.
2. For the PDF in Question 2, determine whether X and Y are independent.

Work in pairs (15 minutes), then we discuss as a class.

Summary and Looking Ahead

Summary: Key Formulas from Today

$$p_{X,Y}(x,y) = P(X = x, Y = y), \qquad f_{X,Y}(x,y):\ P(A) = \iint_A f_{X,Y}\,dx\,dy$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Independent $\iff$ $p_{X,Y} = p_X p_Y$ (or $f_{X,Y} = f_X f_Y$) for *all* x, y

Common Mistakes

- Forgetting that marginal PMF/PDF requires summing/integrating over the **other** variable, not the one of interest
- Treating an iterated integral as new mathematics — it is just two ordinary single-variable integrals
- Checking independence at only **one** point and concluding independence holds everywhere
- Confusing "joint probability is zero" with "variables are independent"
- Forgetting that for non-rectangular regions, integration **limits** may depend on the other variable

Connections Ahead

- Joint distributions (this week) $\rightarrow$ foundation for **covariance** and **correlation** (next week)
- Independence $\rightarrow$ core assumption behind Naive Bayes and many simplified ML models
- Confusion matrices, sensor fusion, and multi-feature datasets are all applications of joint distributions
- Iterated integrals here $\rightarrow$ groundwork for computing expectations of functions of (X, Y), e.g., $E[XY]$

Preview of Next Week: Covariance and Correlation

- Today: whether X and Y are related (independence, yes/no).
- Next week: **how strongly** and in **what direction** they are related.
- We will define $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ and the correlation coefficient ρ.
- Essential groundwork for regression, PCA, and covariance matrices used throughout machine learning.